

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Supplementary Winter Examination – 2023

Course: B. Tech.

Semester : VII

Branch : Electronics and Communication Engineering / Electronics and Telecommunication Engineering

Subject Code & Name: BTETPE703A (Embedded System Design)

Max Marks: 60

Date: 06/01/2024

Duration: 3 Hr.

Instructions to the Students:

- 1. All the questions are compulsory.*
 - 2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.*
 - 3. Use of non-programmable scientific calculators is allowed.*
 - 4. Assume suitable data wherever necessary and mention it clearly.*
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	(Level)	Marks
Q.1 Solve Any Two of the following.		12
A) Define real-time systems and distinguish between hard real-time and soft real-time requirements.	Level 1	6
B) Provide examples of common embedded computing applications and their significance in various industries.	Level 3	6
C) Explore the challenges and opportunities in designing embedded systems for the evolving landscape of IoT.	Level 4	6
Q.2 Solve Any Two of the following.		12
A) Briefly explain classification of the processors in embedded system?	Level 1	6
B) Distinguish between RISC and CISC.	Level 1	6
C) What are the various types of memory in embedded systems?	Level 1	6
Q.3 Solve Any Two of the following.		12
A) Explain interfacing of stepper motor to the microcontroller.	Level 6	6
B) Discuss the significance of signal conditioning in improving the accuracy and reliability of measurements in embedded systems.	Level 4	6
C) Describe the importance of thermal considerations in the design of embedded systems.	Level 2	6
Q.4 Solve Any Two of the following.		12
A) Discuss the characteristics of real-time kernels.	Level 2	6
B) Discuss the advantages and limitations of using polled loops for handling tasks.	Level 2	6
C) Discuss how multi-rate systems manage tasks with varying execution frequencies.	Level 4	6

Q. 5 Solve Any Two of the following.

12

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| A) Explore the advantages of pre-emptive multi-tasking in real-time and responsive systems. | <i>Level 3</i> | 6 |
| B) Explain the concept of Rate-Monotonic Scheduling in real-time systems. | <i>Level 4</i> | 6 |
| C) Discuss the advantages and challenges of using shared memory for communication between tasks. | <i>Level 4</i> | 6 |

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